Auteur: Mathijs Pittoors
Studierichting: Informatica
Oefeningenreeks 2E nr. 10.11

$$
\begin{aligned}
4^{x-1}+1 &= 5\cdot x^{x-2} \\
\frac{4^x}{4}+1 &= \frac{5\cdot 2^x}{4} \\
2^{2x}+4 &= 5\cdot 2^x \\
\textbf{Neem } t &= 2^x \\
t^2-5t+4 &= 0 \\
D &= b^2-4ac = 25-16 = 9 \\
t &= \frac{+5\pm 3}{2} \\
t_1 &= 1 \Rightarrow 2^x = 1 \Rightarrow x = 0 \\
t_2 &= 4 \Rightarrow 2^x = 4 \Rightarrow x = 2
\end{aligned}
$$

References